Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a data scientist, I can gather a historical credit card approval dataset to train the model.	2	High	
Sprint-1		USN-2	As a data scientist, I can load the dataset into a Pandas DataFrame for analysis.	1	High	
Sprint-1	Data Preprocessing	USN-3	As a data scientist, I can handle missing values in the dataset so the model trains on clean data.	3	High	
Sprint-1		USN-4	As a data scientist, I can encode categorical features so the model can process them.	3	High	
Sprint-1		USN-5	As a data scientist, I can scale numerical features for consistent model input.	3	Medium	
Sprint-1		USN-6	As a data scientist, I can remove duplicate and outlier records to improve data quality.	3	Medium	
Sprint-1	Exploratory Data Analysis	USN-7	As a data scientist, I can generate count plots and a correlation heatmap to understand feature relationships.	3	Medium	
Sprint-1		USN-8	As a data scientist, I can split the dataset into training and testing sets.	2	High	
Sprint-2	Model Training	USN-9	As a data scientist, I can train a Logistic Regression model on the processed data.	3	High	
Sprint-2		USN-10	As a data scientist, I can train a Decision Tree	3	High	

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			model on the processed data.			
Sprint-2		USN-11	As a data scientist, I can train a Random Forest model on the processed data.	3	High	
Sprint-2		USN-12	As a data scientist, I can train an XGBoost model on the processed data.	3	High	
Sprint-2	Model Evaluation	USN-13	As a data scientist, I can evaluate all trained models using Accuracy, Precision, Recall, and F1-Score.	5	High	
Sprint-2		USN-14	As a data scientist, I can compare all models and select the best-performing one for deployment.	3	High	
Sprint-3	Flask Application	USN-15	As a developer, I can set up the Flask application structure and routes.	3	High	
Sprint-3		USN-16	As a developer, I can design the prediction form UI using HTML and Bootstrap.	3	High	
Sprint-3		USN-17	As a developer, I can integrate the saved model with the Flask backend to generate predictions.	5	High	
Sprint-3		USN-18	As a developer, I can display the prediction result on a dedicated results page.	2	High	
Sprint-3	Error Handling	USN-19	As a developer, I can validate form input and handle errors gracefully.	3	Medium	
Sprint-3		USN-20	As a developer, I can build a custom 404 page for invalid routes.	2	Low	
Sprint-3	Testing	USN-21	As a QA engineer, I can perform functional testing of the end-to-end prediction flow.	2	High	
Sprint-4	Deployment	USN-22	As a developer, I can deploy the Flask application locally / on IBM Cloud.	5	High	
Sprint-4	Documentation	USN-23	As a technical writer, I can prepare the project report and README documentation.	3	Medium	
Sprint-4		USN-24	As a technical writer, I can prepare the project PPT and demo video.	3	Medium	

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4	Testing	USN-25	As a QA engineer, I can conduct performance and user acceptance testing on the deployed app.	5	High	
Sprint-4	Final Submission	USN-26	As a team, we can finalize the GitHub repository and submit through SmartBridge SkillWallet.	4	High	

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	16 Jun 2026	21 Jun 2026	20	21 Jun 2026
Sprint-2	20	6 Days	23 Jun 2026	28 Jun 2026	20	28 Jun 2026
Sprint-3	20	6 Days	30 Jun 2026	05 Jul 2026		
Sprint-4	20	6 Days	07 Jul 2026	12 Jul 2026		

Velocity:

For the Credit Card Approval Prediction System, the project was planned across 4 sprints of 6 days each, with a total of 80 story points across the full backlog (20 story points per sprint on average). Team velocity = Total Story Points Completed / Number of Sprints = 80 / 4 = 20 story points per sprint. This means the team completes approximately 20 story points of work in every 6-day sprint, which was used to plan the remaining sprints and confirm the project stays on track for the internship deadline.

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts